

MILESTONE 5: Deployment

In Milestone 5, the focus is on deploying the **LegalEase** application locally using **FastAPI** as the backend and **Streamlit** as the frontend. This milestone involves setting up a virtual environment, configuring dependencies, and verifying that all modules—from the Gemini API integration to document generation and downloads—work seamlessly in a real-world local environment. This ensures your system is production-ready and paves the way for cloud deployment in future stages.

Activity 5.1: Preparing the Application for Local Deployment

Step 1: Set Up a Virtual Environment

To ensure environment isolation and consistent dependency management, create and activate a virtual environment:

```
● PS F:\VS CODE\python\LegalEase> venv\scripts\activate
○ (venv) PS F:\VS CODE\python\LegalEase> |
```

Step 2: Install Dependencies

Ensure that the backend (FastAPI) and frontend (Streamlit) dependencies are installed from a requirements.txt:

pip install -r requirements.txt

```
requirements.txt
1  aiohappyeyeballs==2.6.1
2  aiohttp==3.11.16
3  aiosignal==1.3.2
4  altair==5.5.0
5  annotated-types==0.7.0
6  anyio==4.9.0
7  asgiref==3.8.1
8  attrs==25.3.0
9  backoff==2.2.1
10 bcrypt==4.3.0
11 blinker==1.9.0
12 build==1.2.2.post1
13 cachetools==5.5.2
14 certifi==2025.1.31
15 charset-normalizer==3.4.1
16 chroma-hnswlib==0.7.6
17 chromadb==1.0.0
18 click==8.1.8
19 colorama==0.4.6
20 coloredlogs==15.0.1
21 dataclasses-json==0.6.7
```

Step 3: Configure Environment Variables

Create a .env file to securely store your **Gemini API key**

Activity 5.2: Testing and Verifying Local Deployment

Step 1: Launch the FastAPI Backend

Start the backend server using **Uvicorn**:

```
(venv) PS F:\VS CODE\python\LegalEase> uvicorn legalEaseAPI.main:app --reload
INFO:      Will watch for changes in these directories: ['F:\VS CODE\python\LegalEase']
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      Started reloader process [15168] using WatchFiles
INFO:      Started server process [17360]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
```

Step 2: Launch the Streamlit Frontend

Run the app.py file to start the **LegalEase** UI:

```
(venv) PS F:\VS CODE\python\LegalEase> streamlit run frontend/app.py

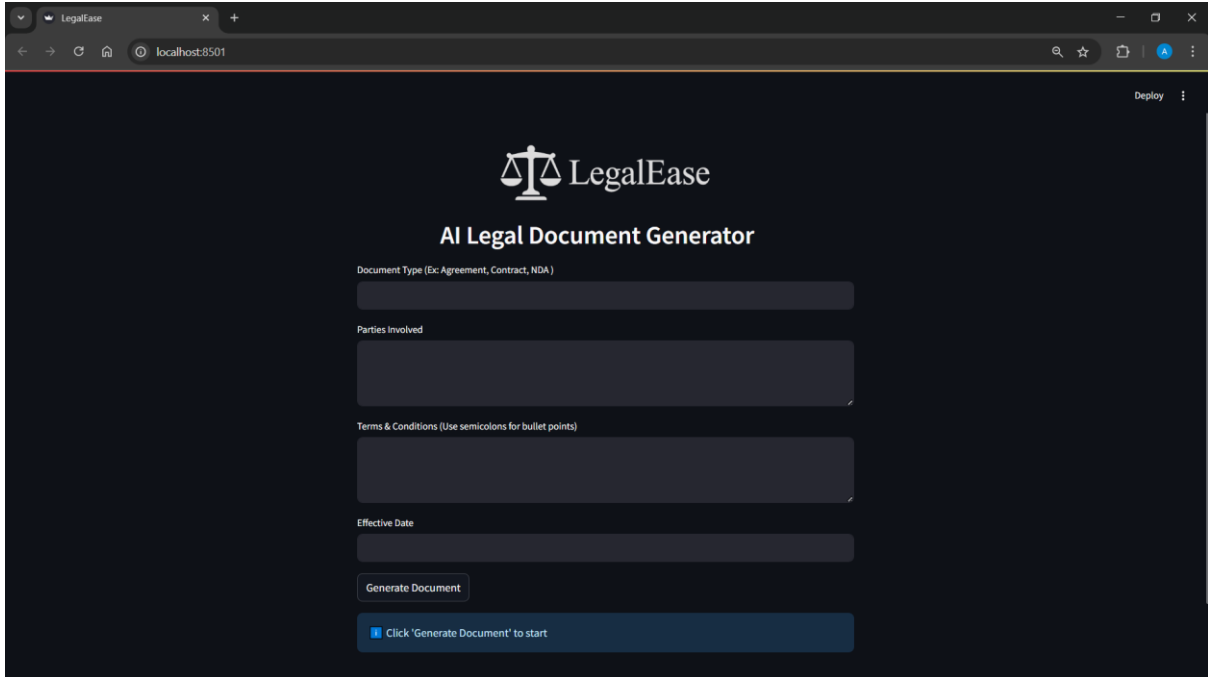
You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.3:8501
```

Step 3: Test All Functionalities

- Input document type, parties, terms, and date.
- Press **Generate Document** to get AI-generated legal content from Gemini.
- Use built-in options to:
 - Preview in dark mode HTML
 - Edit the text
 - Download in .txt, .docx, and .pdf formats
- Check for:
 - Proper layout with logo and footer
 - Clean formatting of bullet points
 - Correct embedding of terms in the document

Exploring LegalEase



LegalEase

AI Legal Document Generator

Document Type (Ex: Agreement, Contract, NDA)

Parties Involved

Terms & Conditions (Use semicolons for bullet points)

Effective Date

Generate Document

Click 'Generate Document' to start

1. Document Type

Purpose: Type of legal document to generate.

Examples:

- Agreement
- Contract
- NDA (Non-Disclosure Agreement)
- Lease Agreement
- Employment Offer Letter

Example input: Freelance Work Contract

2. Parties Involved

Purpose: Names and roles of the individuals or entities involved in the document.

Examples:

- John Doe (Freelancer), ABC Corp (Client)
- Alice Smith (Tenant), XYZ Realty (Landlord)

Example input: Jane Doe (Service Provider), TechNova Inc. (Client)

3. Terms & Conditions (Use semicolons for bullet points)

Purpose: Specific clauses or rules that define the agreement. Separate each bullet point using a semicolon.

Examples:

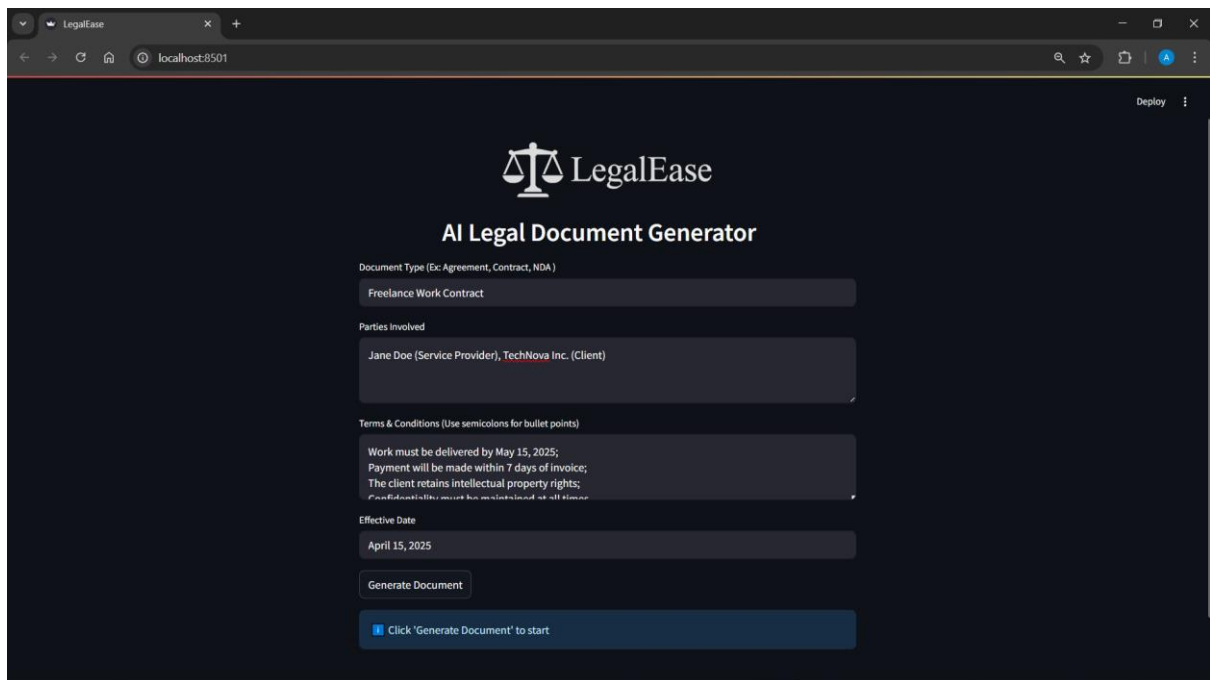
- Payment to be made within 30 days of invoice;
- The provider agrees to deliver work by the agreed deadline;
- Confidentiality must be maintained at all times;
- Either party may terminate with 15 days notice

4. Effective Date

Purpose: The date when the agreement becomes legally valid.

Examples:

- April 10, 2025
- 10/04/2025
- April 15, 2025



The screenshot shows a web browser window with the URL 'localhost:8501'. The application is titled 'LegalEase AI Legal Document Generator'. It features a dark theme with a white logo of a scale of justice. The form includes the following fields:

- Document Type (Ex: Agreement, Contract, NDA):** A dropdown menu with 'Freelance Work Contract' selected.
- Parties Involved:** A text input field containing 'Jane Doe (Service Provider), TechNova Inc. (Client)'.
- Terms & Conditions (Use semicolons for bullet points):** A text input field containing 'Work must be delivered by May 15, 2025; Payment will be made within 7 days of invoice; The client retains intellectual property rights; Confidentiality must be maintained at all times'.
- Effective Date:** A text input field containing 'April 15, 2025'.

Below the form is a 'Generate Document' button and a blue button with a document icon and the text 'Click "Generate Document" to start'.

Document Generated

Effective Date

April 15, 2025

Generate Document

✔ Document Generated Successfully!

Freelance Work Contract

Agreement made this 15th day of April, 2025

Between:

Jane Doe (hereinafter referred to as the "Service Provider"), residing at [Jane Doe's Address], [Jane Doe's City, State, Zip Code]

And:

TechNova Inc. (hereinafter referred to as the "Client"), a corporation organized and existing under

Click to Edit Document

Download as .TXT

Download as .DOCX

Download as .PDF

Editing the Document

If the user wants to edit the document, they can do so by clicking on the “ Click to Edit Document” Button. A new window opens up where the document can be edited.

LegalEase

localhost:8501

Deploy

Do not store personally identifiable information on our site. We will never transfer or sell your personal information to third parties without your consent.

Doe's City, State, Zip Code

And:

TechNova Inc. (hereinafter referred to as the "Client"), a corporation organized and existing under the laws of [State of Incorporation], with its principal place of business at [TechNova Inc.'s Address], [TechNova Inc.'s City, State, Zip Code]

Click to Edit Document

Edit Document Below:

Freelance Work Contract

Agreement made this 15th day of April, 2025

Between:

Jane Doe (hereinafter referred to as the "Service Provider"), residing at [Jane Doe's Address], [Jane Doe's City, State, Zip Code]

And:

TechNova Inc. (hereinafter referred to as the "Client"), a corporation organized and existing under the laws of [State of Incorporation], with its principal place of business at [TechNova Inc.'s Address],

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Edit Document Below:

Freelance Work Contract

I AM EDITING the DOCUMENT|

Agreement made this 15th day of April, 2025

Between:

Jane Doe (hereinafter referred to as the "Service Provider"), residing at [Jane Doe's Address], [Jane Doe's City, State, Zip Code]

And:

Press Ctrl+Enter to apply

Downloading/ Saving the Generated Document

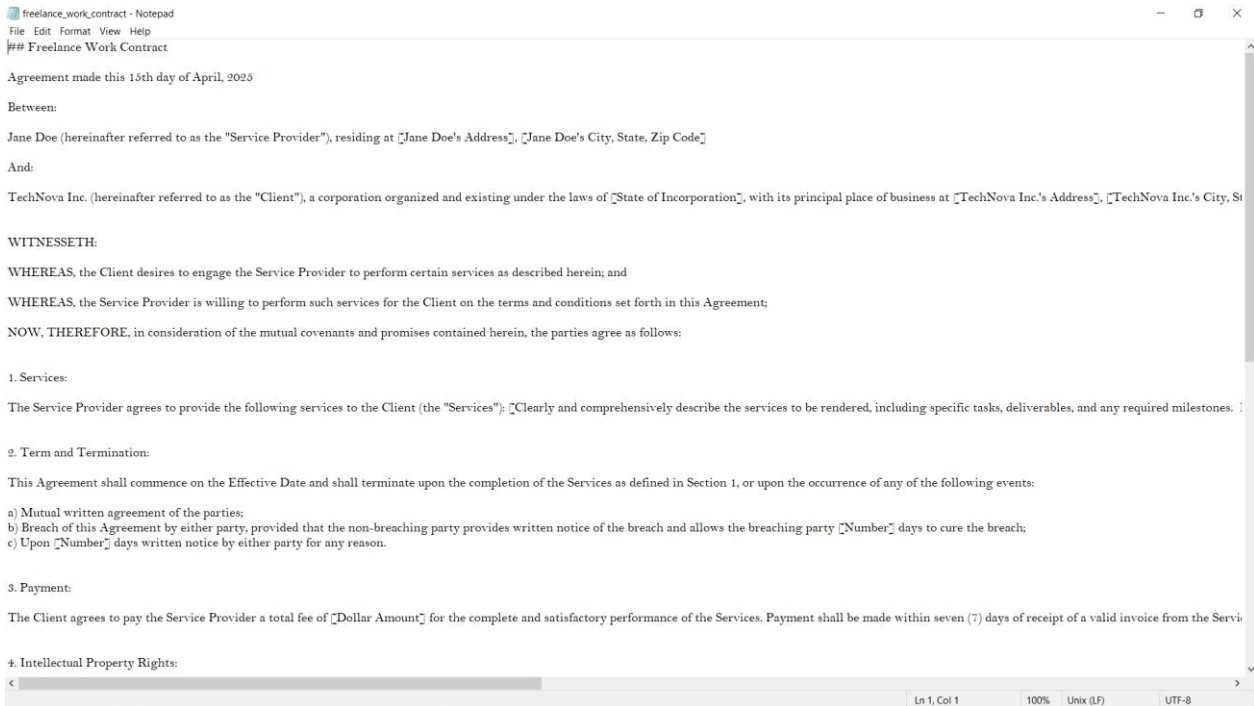
We can Download and save the documents in any of the following formats.

 Download as .TXT

 Download as .DOCX

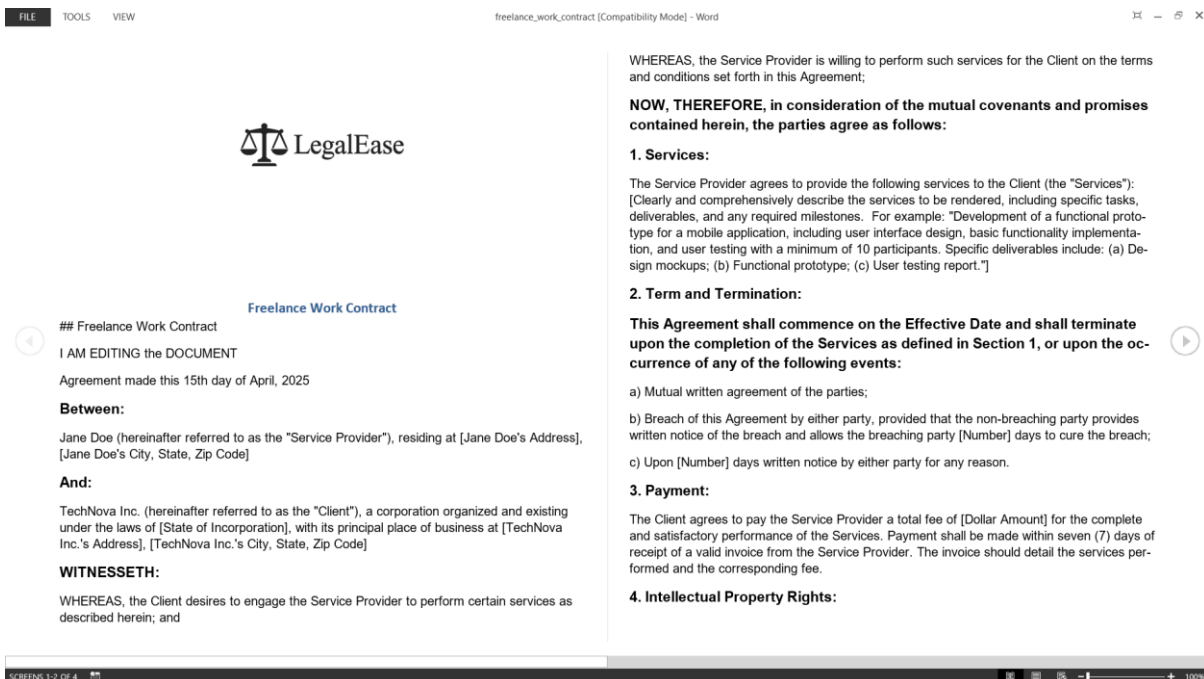
 Download as .PDF

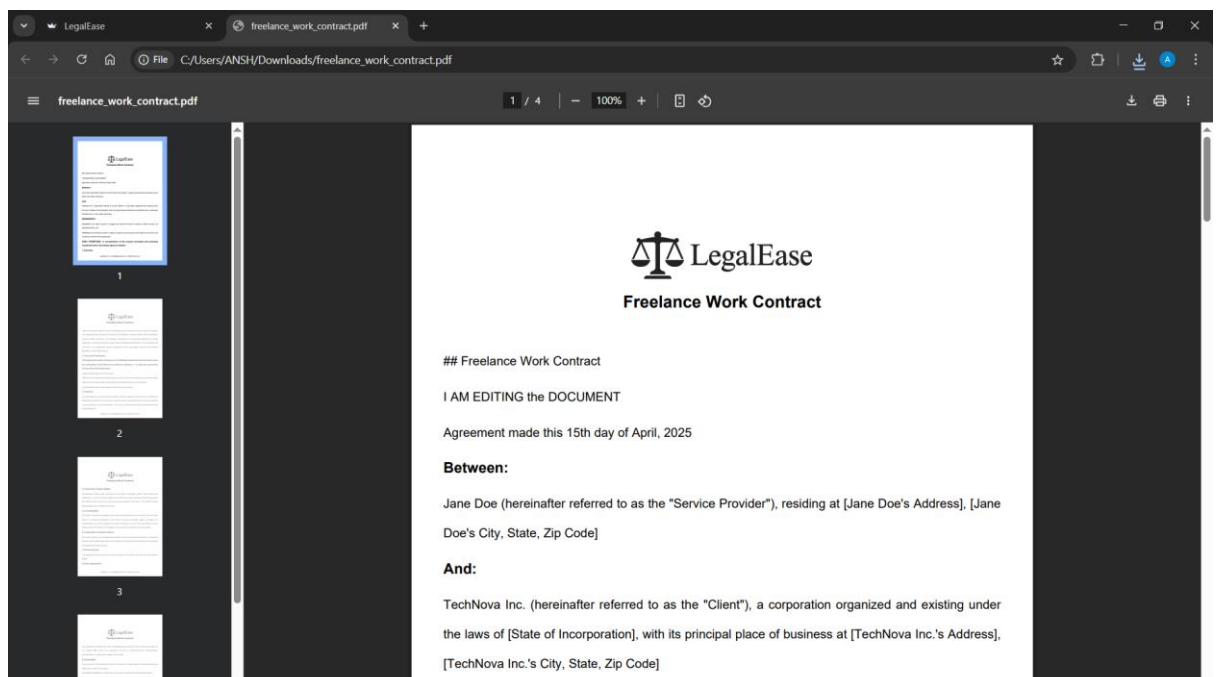
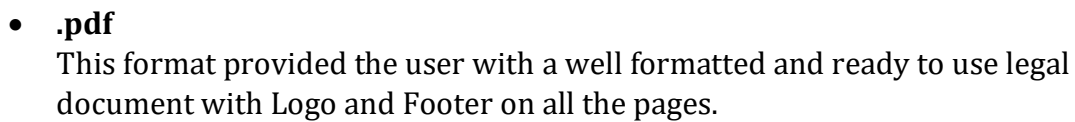
- .txt

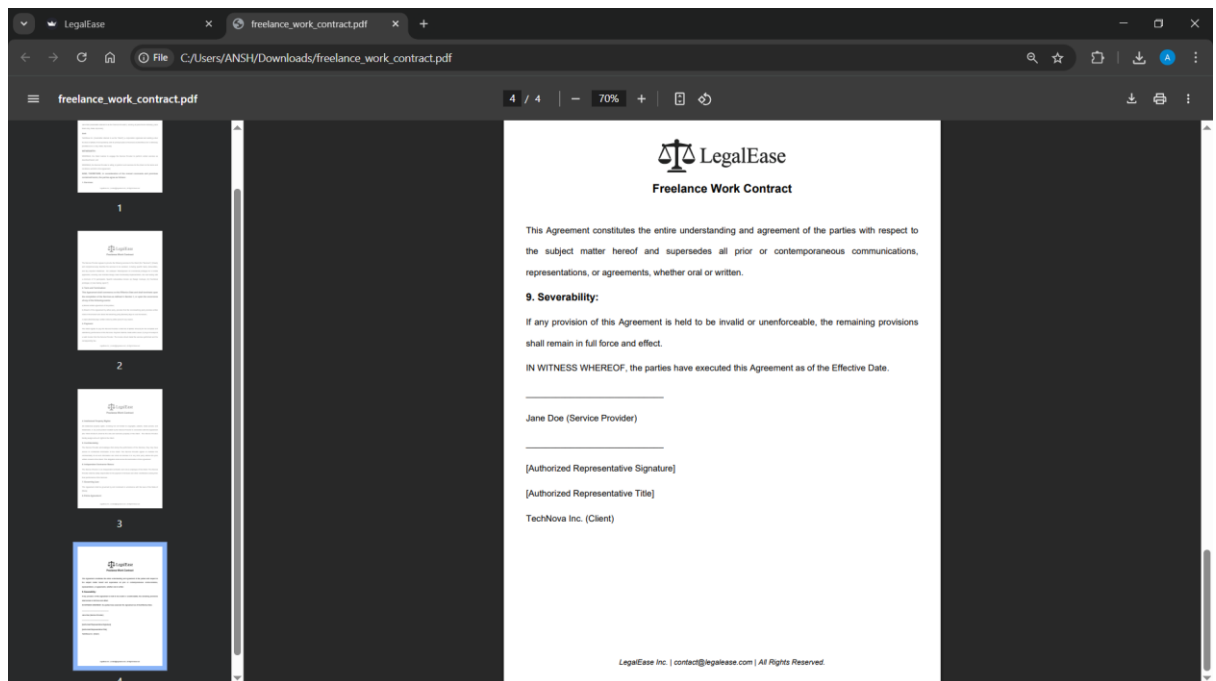


- **.docx**

The document provided inculcates the use of the logo (front page) as well as the footer in the last page







Conclusion

LegalEase represents a transformative step in bridging the accessibility gap in legal services through the power of artificial intelligence. This project demonstrates how emerging technologies can simplify complex legal language, providing users—regardless of legal background—with a clear, concise understanding of legal documents. By integrating advanced natural language processing (NLP) models with user-centric design, LegalEase empowers individuals to navigate legal content with confidence, thereby promoting inclusivity and informed decision-making.

The platform's ability to analyze and simplify legal text, highlight key clauses, and provide plain-language summaries enhances comprehension and reduces dependency on costly legal consultations for basic understanding. This democratization of legal knowledge is critical in a world where legal literacy remains low, especially among marginalized populations. Furthermore, LegalEase supports multilingual capabilities, ensuring broader reach and utility across different regions and demographics.

From a technical standpoint, LegalEase showcases efficient integration of backend AI models, including document parsing, summarization algorithms, and user-friendly web interfaces. The project's architecture is designed to ensure scalability, data privacy, and real-time responsiveness—key elements for any modern SaaS solution. Through iterative development, testing, and feedback, the system has been refined for optimal performance and user satisfaction.

In conclusion, LegalEase is more than a technological tool—it is a catalyst for legal empowerment. By fostering transparency, accessibility, and understanding, it upholds the principle that legal information should be a public good, not a privileged resource. Future enhancements could include deeper contract analysis, integration with legal databases, and personalized recommendations. With continued innovation and support, LegalEase has the potential to revolutionize how people interact with the law, making justice more approachable for all.